

Calculus 2 - Homework 3 (Worked Solutions)

1.

Both an improper integral and a parts problem. First find the antiderivative by parts: $u = x$, $dv = e^x dx \rightarrow du = dx$, $v = e^x$.
 $\int x e^x dx = x e^x - \int e^x dx = x e^x - e^x$.

Now take the limit: $\int_0^t x e^x dx = [x e^x - e^x]_0^t$

$$= (t e^t - e^t) - (0 - 1) = 1 - t e^t + e^t.$$

As $t \rightarrow \infty$, both $t e^t \rightarrow \infty$ and $e^t \rightarrow \infty$, so the limit is ∞ .

ANSWER: converges to ∞

2.

It is tempting to apply the Alternating Series Test, but first check the necessary condition that the terms go to 0.

Here $a_n = (-1)^n \frac{n}{n+1}$. The magnitude $b_n = \frac{n}{n+1} \rightarrow 1$, not 0, as $n \rightarrow \infty$. So a_n does not approach 0.

By the n th-Term Divergence Test, the series diverges (the Alternating Series Test never applies because its limit condition fails).

ANSWER: diverges (n th-term test: terms do not approach 0)

3.

Alternating Series Test: $b_n = \frac{1}{\sqrt{n}}$ is positive, decreasing, and $\lim b_n = 0$, so the series converges.

Absolute values: $\sum \frac{1}{\sqrt{n}} = \sum \frac{1}{n^{1/2}}$ is a p -series with $p = 1/2 < 1$, which diverges.

Converges but not absolutely $\rightarrow$ conditionally convergent.

ANSWER: conditionally convergent

4.

Even power of sine: use the power-reduction identity

$$\sin^2 x = \frac{1 - \cos 2x}{2}.$$

$$\int \sin^2 x dx = \int \frac{1 - \cos 2x}{2} dx$$

$$= \frac{1}{2} \int 1 dx - \frac{1}{2} \int \cos 2x dx$$

$$= \frac{1}{2} x - \frac{1}{2} \left(\frac{\sin 2x}{2} \right) + C = \frac{x}{2} - \frac{\sin(2x)}{4} + C.$$

ANSWER: $\frac{x}{2} - \frac{\sin(2x)}{4} + C$

5.

$\Delta x = 0.25$. Nodes: 0, 0.25, 0.5, 0.75, 1.

$$f(x) = \frac{1}{1+x^2}: f_0=1, f_1=1/1.0625=0.941176, f_2=1/1.25=0.8,$$

$$f_3=1/1.5625=0.64, f_4=1/2=0.5.$$

$$S = \frac{0.25}{3} [f_0 + 4 f_1 + 2 f_2 + 4 f_3 + f_4]$$

$$= 0.083333 [1 + 4(0.941176) + 2(0.8) + 4(0.64) + 0.5]$$

$$= 0.083333 [1 + 3.764706 + 1.6 + 2.56 + 0.5] = 0.083333(9.424706) = 0.785392.$$

(Exact $\pi/4 = 0.785398$; Simpson is accurate to 5 decimals here.)

ANSWER: $S_4 = 0.785392$

6.

The rational form might invite the integral test, but its antiderivative is messy; limit comparison is far cleaner. For large n the term behaves like $\frac{n}{n^3} = \frac{1}{n^2}$, so compare with $b_n = \frac{1}{n^2}$.

$$\lim_{n \rightarrow \infty} \frac{(n+2)/(n^3+n)}{1/n^2}$$

$$= \lim_{n \rightarrow \infty} \frac{n^2(n+2)/(n^3+n)}{1} = \lim_{n \rightarrow \infty} \frac{n^3 + 2n^2}{n^3 + n} = 1 \text{ (finite, positive).}$$

Since $\sum \frac{1}{n^2}$ converges, so does $\sum \frac{n+2}{n^3+n}$.

ANSWER: converges (limit comparison with $\sum 1/n^2$, ratio 1)

7.

For an alternating series meeting the Leibniz conditions, the error after N terms is at most the first omitted term: $|R_N| \leq b_{(N+1)} = 1/(N+1)^3$.

Require $1/(N+1)^3 < 0.01$, i.e. $(N+1)^3 > 100$.

Since $4^3 = 64 < 100$ and $5^3 = 125 > 100$, we need $N + 1 \geq 5$, so $N \geq 4$.

The smallest number of terms is $N = 4$.

ANSWER: $N = 4$ terms

8.

$\Delta x = (2 - 1)/4 = 0.25$. Nodes: 1, 1.25, 1.5, 1.75, 2.

$f(x)=1/x$: $f_0=1$, $f_1=0.8$, $f_2=0.6667$, $f_3=0.5714$, $f_4=0.5$.

$T = (0.25/2)[f_0 + 2f_1 + 2f_2 + 2f_3 + f_4]$

$= 0.125[1 + 2(0.8) + 2(0.6667) + 2(0.5714) + 0.5]$

$= 0.125[1 + 1.6 + 1.3334 + 1.1428 + 0.5] = 0.125(5.5762) = 0.69703$.

(Exact value $\ln 2 = 0.69315$; trapezoid overestimates a convex integrand.)

ANSWER: $T_4 = 0.69703$

9.

$a_n = n!/n^n$. Ratio:

$a_{(n+1)}/a_n = [(n+1)!/(n+1)^{(n+1)}] * [n^n/n!]$

$= (n+1) * n^n/(n+1)^{(n+1)} = n^n/(n+1)^n = (n/(n+1))^n = 1/((n+1)/n)^n$

$= 1/(1 + 1/n)^n$.

As $n \rightarrow \infty$, $(1 + 1/n)^n \rightarrow e$, so $L = 1/e$.

Since $L = 1/e$ is about $0.368 < 1$, the series converges by the Ratio Test.

ANSWER: converges (ratio test, $L = 1/e < 1$)

10.

Form $\sqrt{a^2 - x^2}$ with $a = 2$. Substitute $x = 2 \sin \theta$,

$dx = 2 \cos \theta d\theta$, $\sqrt{4 - x^2} = 2 \cos \theta$, $x^2 = 4 \sin^2 \theta$.

$\int (4 \sin^2 \theta)/(2 \cos \theta) * 2 \cos \theta d\theta = 4 \int \sin^2 \theta d\theta$.

Power-reduce: $4 \int (1 - \cos 2\theta)/2 d\theta = 2(\theta - \sin(2\theta)/2)$

$= 2\theta - \sin(2\theta) = 2\theta - 2 \sin \theta \cos \theta$.

With $\sin \theta = x/2$, $\cos \theta = \sqrt{4 - x^2}/2$, $\theta = \arcsin(x/2)$:

$\sin \theta \cos \theta = x \sqrt{4 - x^2}/4$, so $2 \sin \theta \cos \theta = x \sqrt{4 - x^2}/2$.

Result $= 2 \arcsin(x/2) - (x/2) \sqrt{4 - x^2} + C$.

ANSWER: $2 \arcsin(x/2) - (x/2) \sqrt{4 - x^2} + C$